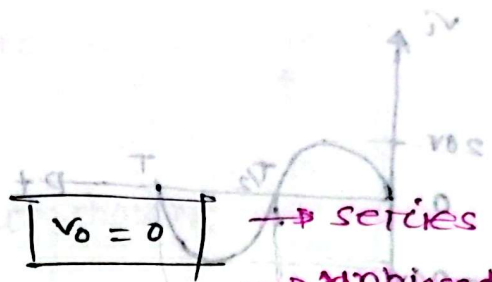
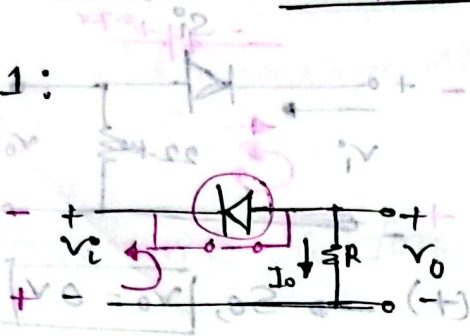
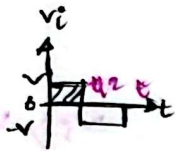
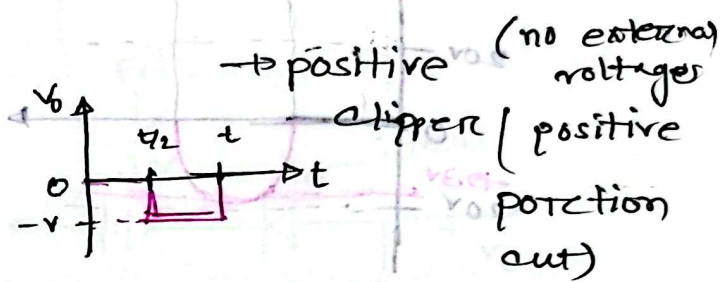


Clippers

Ex. 1:

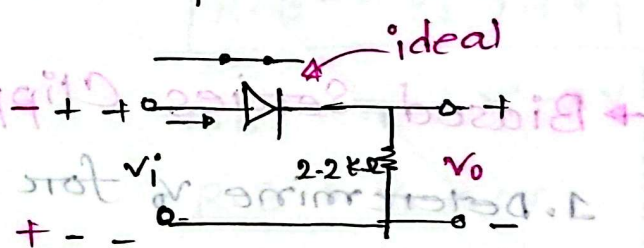
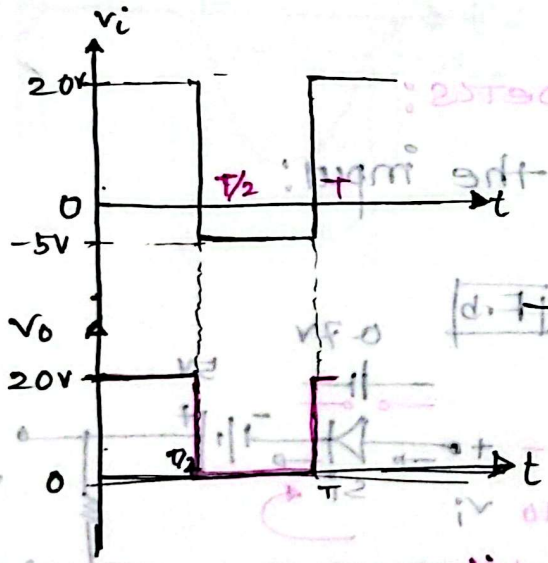


negative : Forwarded
 $-v_i + v_o = 0$
 $\Rightarrow v_o = v_i$
 $\therefore v_o = -V$



unbiased Series Clippers:

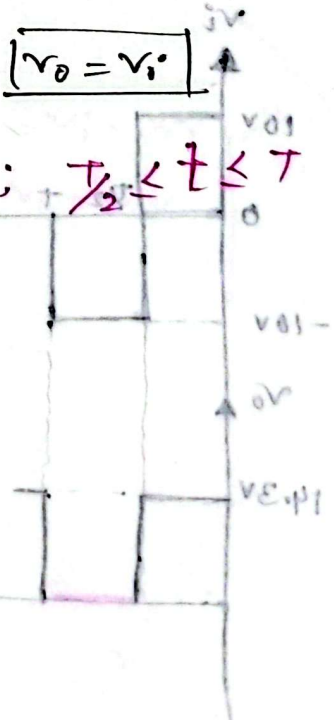
1. Determine v_o for the input shown;



$v_i = 20V$; $v_o = v_i$

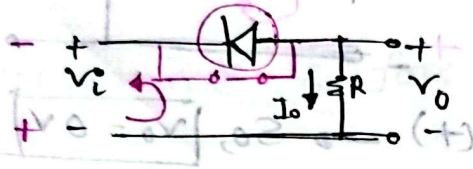
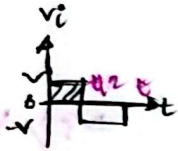
$v_o = 0$; $T/2 \leq t \leq T$

negative clipper



clippers

Ex. 1:



$$v_o = 0$$

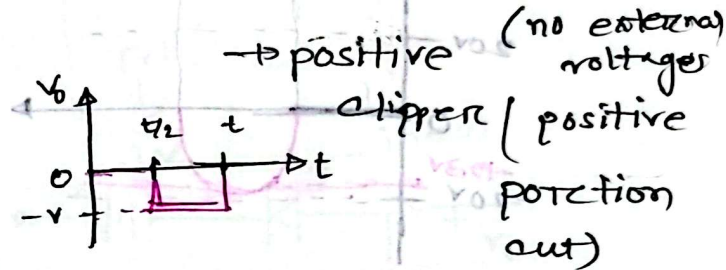
→ series clipper
→ unbiased

negative Forward

$$-v_i + v_o = 0$$

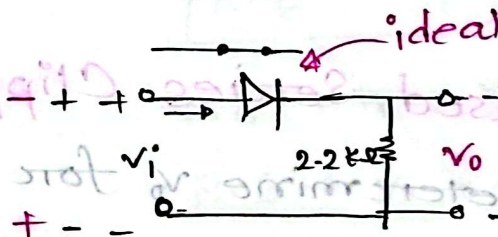
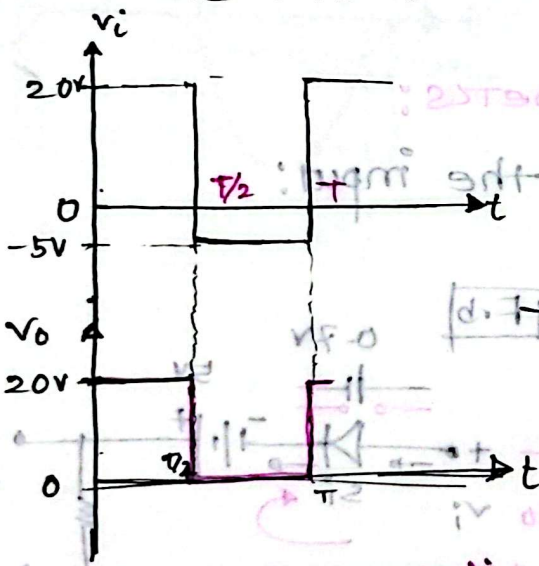
$$\Rightarrow v_o = v_i$$

$$\therefore v_o = -V$$



unbiased Series Clippers:

1. Determine v_o for the input shown;

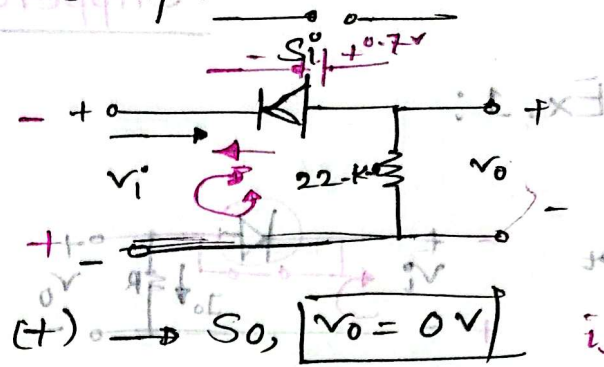
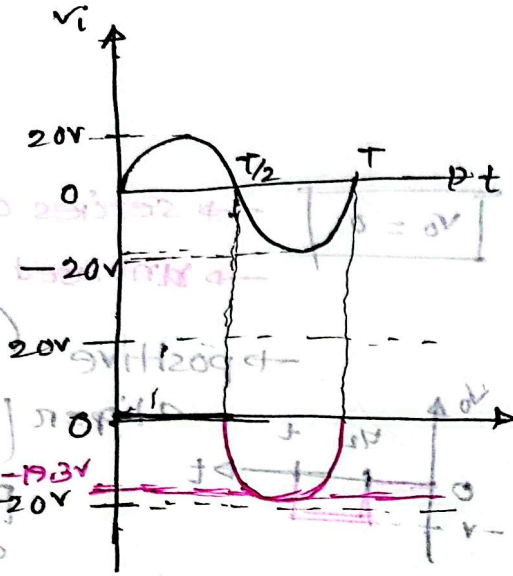


$$v_i = 20V ; v_o = v_i$$

$$\rightarrow v_o = 0 ; T/2 \leq t \leq T$$

→ negative clipper

2. Determine v_o for the input shown:



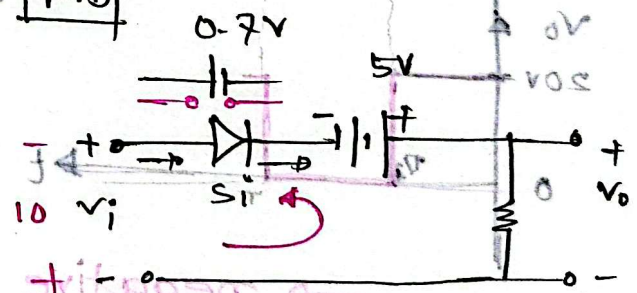
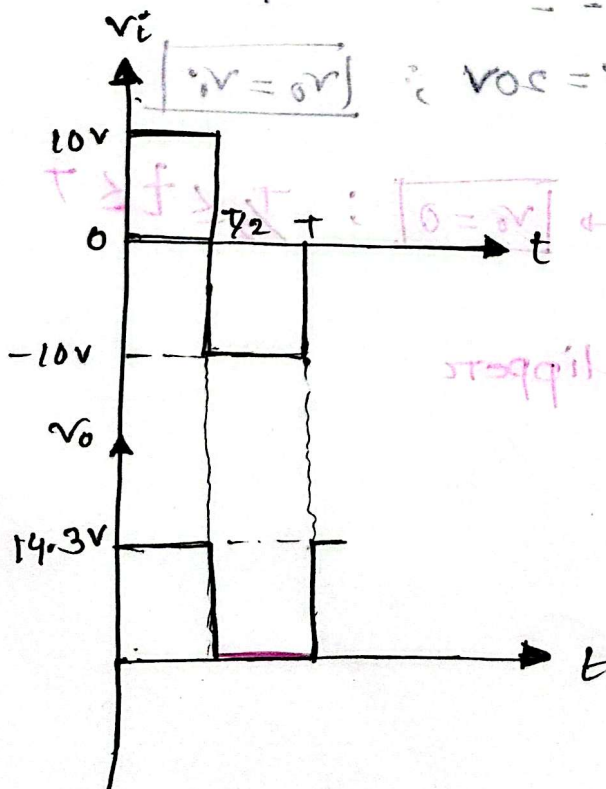
For (+) $v_i + 0.7V = v_o$
 $0 = 0V + 0.7V$
 $v_o = v_i + 0.7V$
 $v_i = 0V$
 $= -20 + 0.7V$
 $= -19.3$

$v_i + v_o - 0.7 = 0$
 $v_o = -v_i + 0.7V$

positive clipper

→ Biased Series Clippers:

1. Determine v_o for the input:



state of diode: - Forw. Rev.

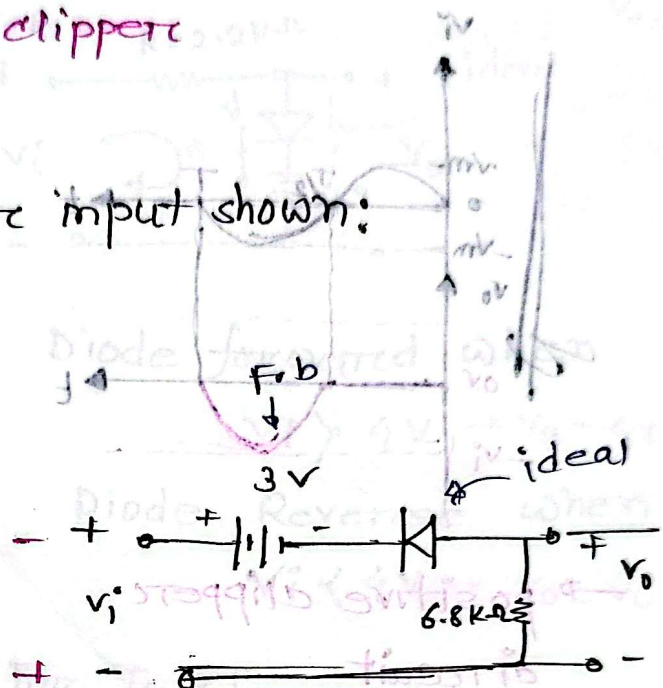
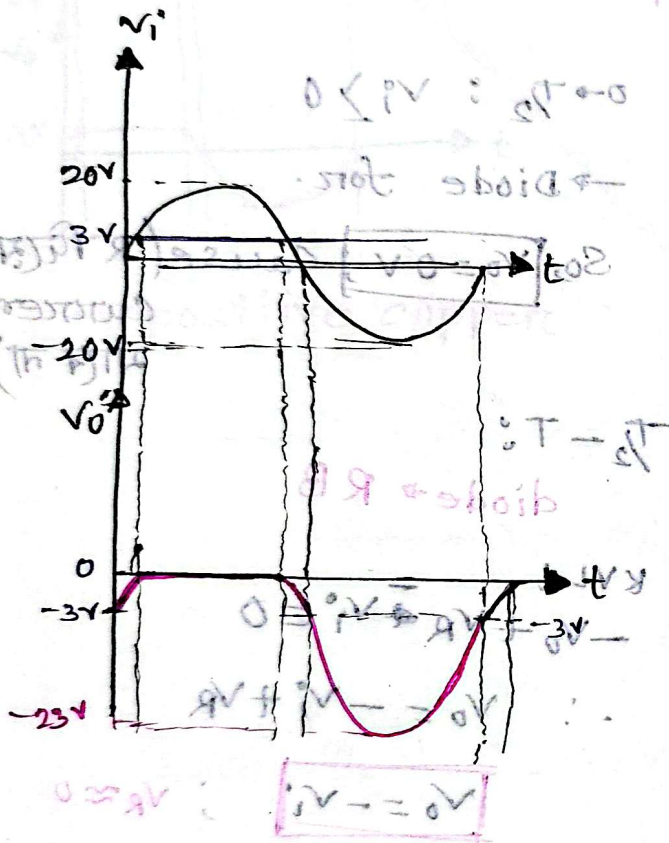
For (+) Forw:

$v_o = v_i - 0.7V + 5V$
 $= 10 - 0.7 + 5$
 $\therefore v_o = 14.3V$

For (-) : R:B || $v_o = 0V$

→ negative clipper

2. Determine v_o for input shown:



$$\begin{aligned} F.B &\rightarrow v_i < 3V \\ R.B &\rightarrow v_i > 3V \end{aligned}$$

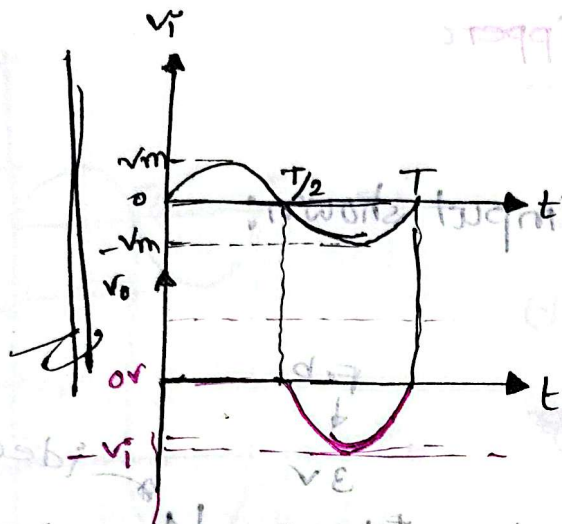
$$\begin{aligned} F.B &\rightarrow v_o = v_i - 3V \\ R.B &\rightarrow v_o = 0V \end{aligned}$$

$$\begin{aligned} v_i &= 0 \\ v &= -3V \\ v_i &= 3V \\ v_o &= 0 \end{aligned}$$

$$\begin{aligned} v_o &= -v_i - 3V \\ v_i &= 0 \Rightarrow v_o = -3V \end{aligned}$$

$$\begin{aligned} v_o &= -20 - 3 \\ &= -23 \end{aligned}$$

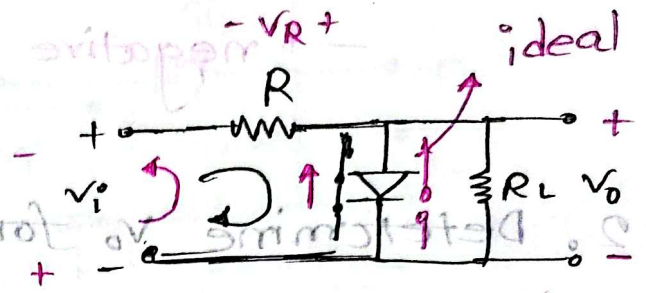
unbiased parallel clipper:



→ positive clipper circuit

$$\begin{aligned} V_E > V &\rightarrow \text{off} \\ V_E < V &\rightarrow \text{on} \end{aligned}$$

$$\begin{aligned} V_E - V &= 0 \rightarrow \text{off} \\ V_E - V &= 0 \rightarrow \text{on} \\ V_E - V &= 0 \rightarrow \text{on} \end{aligned}$$



$$0 \rightarrow T/2 : V_i \geq 0$$

→ Diode forw.

So, $V_o = 0V$ cause (R diya current ma k na)

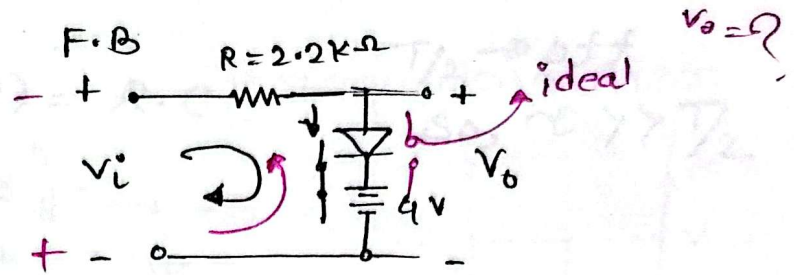
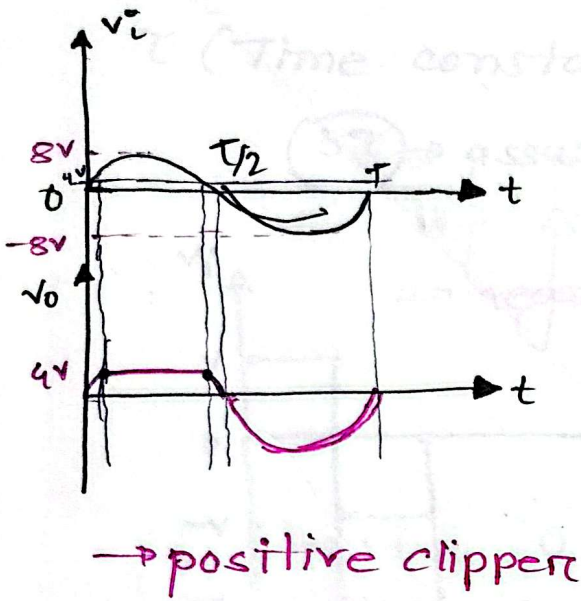
$$T/2 - T :$$

diode → RB

$$\begin{aligned} \text{KVL} \\ -V_o + V_R + V_i &= 0 \\ \therefore V_o &= -V_i + V_R \end{aligned}$$

$$\boxed{V_o = -V_i} ; V_R \approx 0$$

Biased Parallel Circuits



Diode forward when

$$v_i > 4V \rightarrow v_o = 4V$$

Diode Reverse when

$$v_i < 4V \rightarrow v_o = v_i / v_o - v_i = 0$$

For $T/2 \rightarrow T$:

$$R_{eq} = 0; S_{eq} = 0; \tau = 0$$

$$+v_i - v_o = 0; v_o = v_i = \pm V$$

$$\text{From } T/2 \text{ to } T: v_i + v_o + V = 0$$

$$v_o = -v_i - V = -2V$$

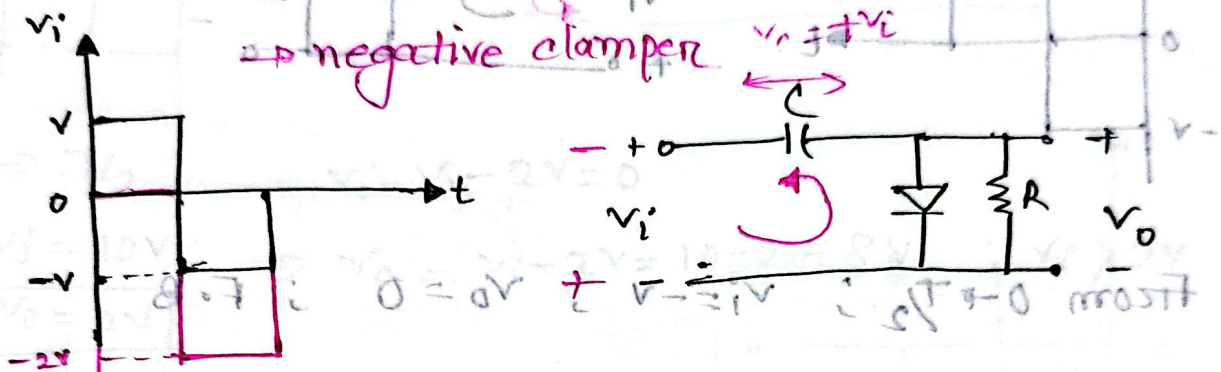
i) rms remain same

ii) peak value changes.

Positive and negative clamper

τ (Time constant) = $R \cdot C$ $T/2 \rightarrow \text{off}$
So, $\tau \gg T/2$

(5) τ assume



0 to $T/2$; $V_i = \pm V$ and $V_o = 0$; f.B

Res = 0; So, $\tau = 0$

$+V_i - V_c = 0$; $V_c = V_i = \pm V$

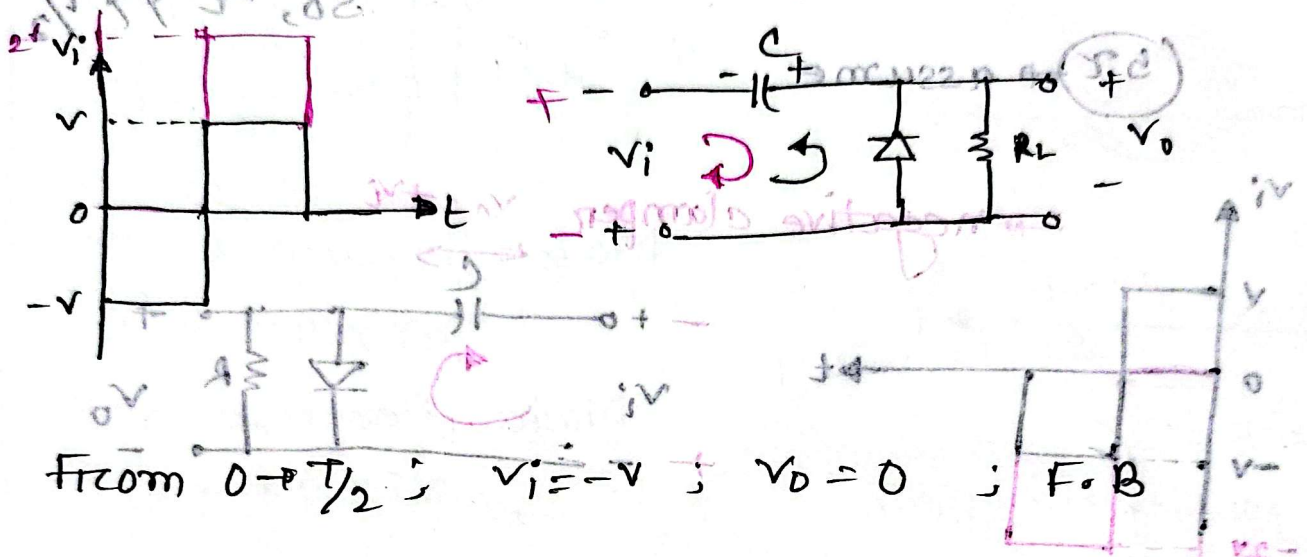
From $T/2$ to T ; $V_i + V_o + V_c = 0$; R.B

$V_o = -V_i - V = -2V$

- * i) rms remain same
- ii) peak value changes.

positive and negative clamper circuits:

positive clamper:



$$0 = 0V + V - V_c = 0 \Rightarrow V_c = V = -(-V_i)$$

$$V_c = V = -(-V_i)$$

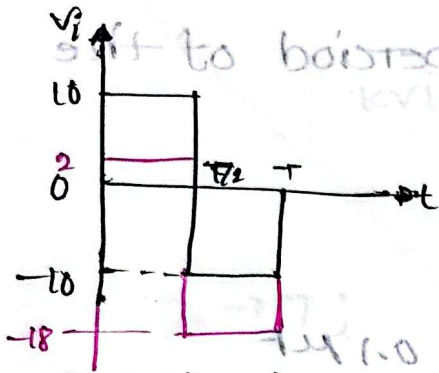
From $T/2 \rightarrow T$; R.B $V_o = V_i + V_c$ (KVL)

$$0 = V + 0V + V_c = 0 \Rightarrow V_o = V + V = 2V$$

$$V_o = 2V$$

- i) same waveform same
- ii) peak value changes.

Biased clamper circuit: Additional voltage swing



$$0 \Rightarrow T/2; \quad v_i - v_c - 2v = 0$$

$$V_i = 10 \text{ V}$$

$$\Rightarrow V_C = V_i - 2V = 10 - 2 = 8V \quad ; \quad V_i > 2V$$

$$V_0 = 2V$$

$$T_2 \rightarrow T;$$

Applying

KVL -

$$V_i + V_o + V_c = 0$$

$$\Rightarrow v_0 = -v_1 - v_c$$

$$(a) \quad 2 \times 10^2 \times 10^3 \times 10^4 \times 10^5 = 2 \times 10^{14}$$

$$= -18 \text{ V}$$

$$2m\sqrt{2} =$$

$$\therefore P_A = P \times P \cdot e = 58 \text{ m sec}$$

$$0.2 \pi \times 10 = \frac{0.2 \pi \times 1}{5} = \frac{1}{5}$$

$$25 = 5 \times 5$$

15/11/2015

$$22 = \frac{28}{0.2} = \frac{56}{1}$$

$$(c/r) \times \sqrt{2} = \sqrt{2} \therefore$$

$$2 \epsilon_0 = \frac{1}{\epsilon_0 \times 1} = \frac{1}{t} = T \quad (d)$$

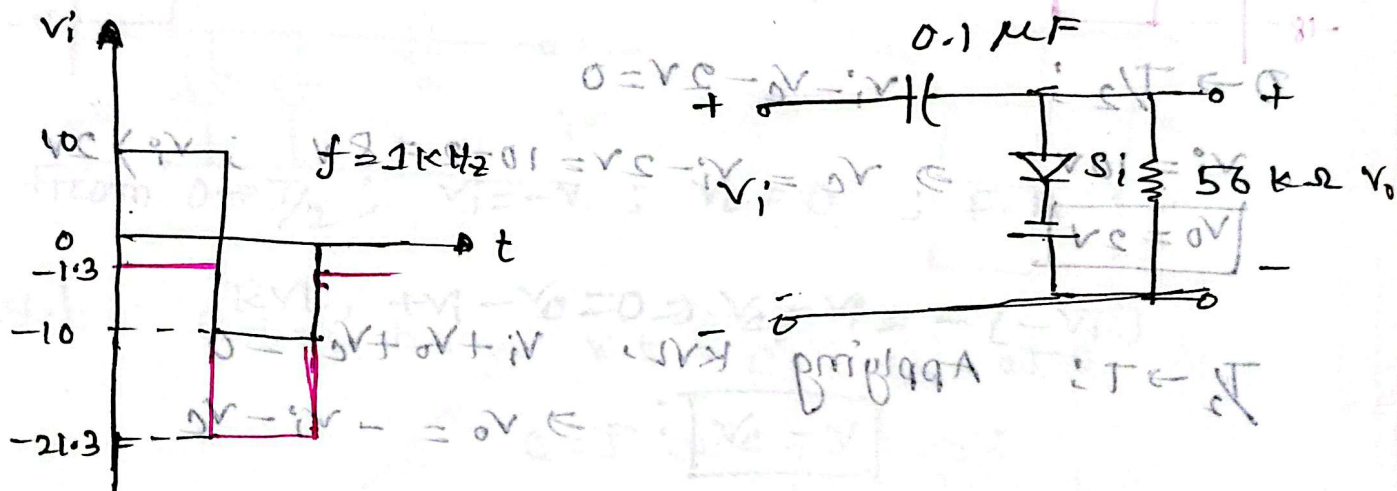
$$552 \text{ m L} = 2 \text{ L}$$

$$v_0^2 = v_1^2 + v_2^2$$

problem: Additional: Question: frequency based

- calculate 5τ
- compare 5τ to half the period of the applied signal.

c) Sketch V_o



$$a) \tau = RC = 56 \times 10^3 \times 0.1 \times 10^{-6} \text{ s} = 5.6 \text{ ms}$$

$$\therefore 5\tau = 5 \times 5.6 = 28 \text{ mSec}$$

$$b) T = \frac{1}{f} = \frac{1}{1 \times 10^3} = 10^{-3} \text{ s}$$

$$T = 1 \text{ mSec}$$

$$\therefore T/2 = 0.5 \text{ mSec}$$

$$T/2 = \frac{1 \text{ mSec}}{2} = 0.5 \text{ mSec}$$

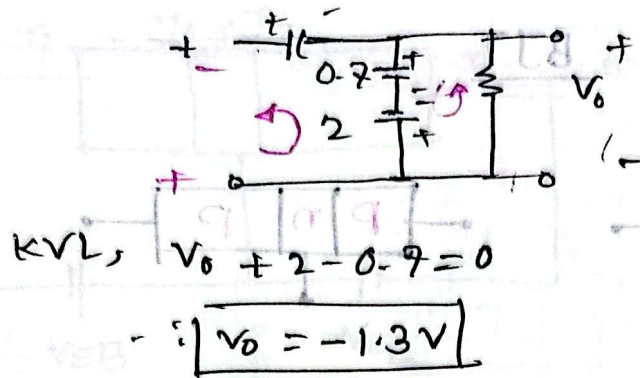
$$5\tau = 28 \text{ mSec}$$

$$\therefore 5\tau \gg T/2$$

$$\frac{5\tau}{T/2} = \frac{28}{0.5} = 56$$

$$\therefore 5\tau = 56 \times (T/2)$$

c) $0 \rightarrow T/2$;



F-b

$$V_i - V_o - 0.7 + 2 = 0$$

$$\Rightarrow V_o = V_i - 0.7 + 2$$

$$= 10 - 0.7 + 2$$

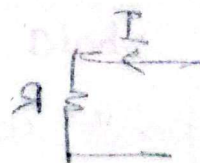
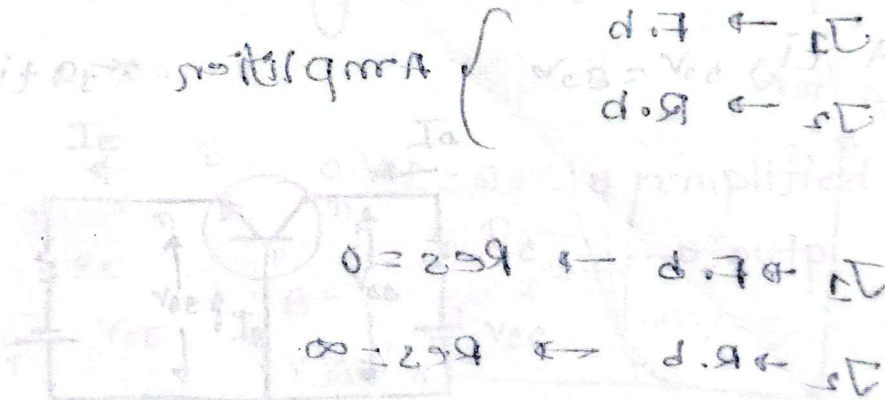
$$V_o = 11.3V$$

$T/2 \rightarrow T$;

$$V_i = -10V$$

$$V_i + V_o + V_c = 0$$

$$\Rightarrow V_o = -V_i - V_c = -(-10) - 11.3 = 10 - 11.3 = -1.3V$$



$$I = 2mA$$



$$I = 2mA$$

